# Wayfinder

**A robot guide for people losing their sight: you lead whenever you can; it reads how you are doing through the handle and steps in only when you need it, and says why.**

AgentVerse 2026 · Robotics track · Anusha Upadhyay (solo) · simulation: MuJoCo, Python, runs on a laptop CPU

## USER AND PROBLEM

**Who:** people who have recently lost their sight, mostly older adults with glaucoma or macular degeneration, in the gap between diagnosis and help. Nearly 80% of the UK's 2M+ people with sight loss are 65 or older [1].

**The gap:** after certification, 85% of English councils miss the 28-day target for rehab assessment [2]; 980 people are waiting for a guide dog, many for over two years, and dogs mostly go to a younger minority [3]. At stations, only 44% of blind passengers are reliably met by booked assistance [4].

*"When I got lost, I was too scared to go out – I lost all my confidence." Jennie [3]. "I know the assistance is going to fail. No one meets me." Paul Goddard [4].*

The pain is lost confidence, which keeps people indoors. A glaucoma specialist told us her patients talk most about losing independence (personal communication).

## KEY FEATURES (DELIBERATELY FEW)

- **Reads the person, not just the room.** Grip force and how the user follows on the handle give a tension estimate, calibrated to each person (works with weak, arthritic grips). Tense: it slows and describes what's around. Calm: it talks less.
- **You stay in charge; control moves both ways.** Calm on a known route: the user steers and Wayfinder only watches out. Tense, busy or near danger: it takes over and says why. Sight loss is permanent and confidence varies day to day, so it adapts in both directions rather than "graduating" anyone. Users in guide studies wanted control, not to be dragged.
- **Remembers regular routes and says what has changed** ("something new, 4 metres ahead on your left"). Blind people memorise their routes and RNIB says changes must be communicated accessibly; 56% name temporary obstacles as a barrier [6].
- **Never steps the user into a drop.** A separate safety layer refuses unsafe moves, like a guide dog's intelligent disobedience, and explains the refusal.
- **Reads the crowd 2 s ahead** and only avoids places people will be when it gets there; commits to a side instead of dithering.

Hardware notes: heated handle so users keep gloves off in winter and grip sensing stays accurate.

## SYSTEM ARCHITECTURE

**Sense:** one forward depth camera (160×120, 25 Hz) + a ring of 9 short-range distance sensors + handle grip.
**Perceive:** each depth pixel becomes a 3D point labelled floor / obstacle / missing floor (below floor level).
**Map:** 25 cm grid; walls and pillars known in advance, everything else learned live; drops are never forgotten; route memory saved per journey.
**Track and predict:** socially aware navigation. Moving blobs are tracked; constant-velocity trajectory prediction 2 s ahead (a strong baseline [7]; the user-tested CaBot system used 4 s [8]), with uncertainty growing over time and with how erratic each person is; avoiding the "freezing robot problem" [9].
**Plan:** A* with safety margins (50 cm obstacles, 85 cm drops, predicted people), re-planned 2–4× a second; switching sides costs extra.
**Decide:** lead/shadow policy from familiarity + user tension; independent safety layer can always override.
**Act and speak:** steering plus on-device speech, one line at a time like a person; only safety warnings cut in.

## WHERE IT RUNS, AND HOW FAST

| PART | WHERE | TIME |
|---|---|---|
| Depth labelling + mapping | CPU | <1 ms / frame |
| Tracking + 2 s prediction | CPU | <1 ms / frame |
| A* re-plan (80×40 grid) | CPU | ≈13 ms, 2–4×/s |
| Speech | on-device TTS | no network |

**Stated budget:** one CPU core, a 40 ms control cycle (25 Hz), no GPU, no network. Measured on a laptop CPU: typical decision step under 1 ms, worst-case re-plan about 13 ms (a third of the budget); the whole simulation including camera rendering runs about 5× faster than real time.

## EVIDENCE (RANDOM CROWDS AND EVENTS)

Prediction horizon, same 20 random crowds: no prediction, 2 s and 4 s all had 0 contacts, because the safety layer holds the distance on its own. 4 s stopped 35% more often than 2 s and arrived 7% later, so we use 2 s; its job is to yield early instead of freezing.

Journey 1 (new route), 40 random runs: all 40 arrived, 0 times the robot moved into anyone, never closer than 0.43 m to a person or 0.78 m to the drop. Journey 2 (remembered route), 40 runs: 39 arrived, 0 robot-caused contacts, never closer than 0.72 m to the drop. It handed the lead to the user in 34 of 40 runs, only once the drop was behind them, and had to take back control from a step toward the drop in 3.

## CRITICAL FAILURE MODE AND SAFETY

**Leading someone off an edge** (platform edge, open hatch, stairs down). Visually impaired people are 9–15% of platform falls with no train present, against ~3% of the population [5]. Wayfinder labels missing floor directly from depth, keeps an 85 cm margin, never forgets a drop, and a safety layer stops it whenever missing floor is within 1.4 m ahead, whatever the planner or the user wants. In shadow mode it takes over and says "Stop. There's a drop that way. Bear left." If the camera loses most of its picture, or there is no safe route, it stops and says so. If the user lets go of the handle, it stops.

## LIMITS

- Simulation only: perfect sensors, no wheel slip. Real depth cameras fail on glass and shiny floors; a real build adds close-range sensing.
- Tension is a modelled handle signal; it needs validating against real users.
- Constant-velocity prediction; a Kalman filter or learned predictor is the next step.
- Indoor concourse only. Outdoor kerbs, crossings and stairs are future work.
- Not yet tested with blind users. Next: co-design sessions, as Guide Dogs already runs with tech companies.

## WHAT IS NEW

Robot guides already avoid obstacles and predict crowds (CMU/IBM CaBot, 2020; Glidance Glide). Grip force has been sensed and stress in blind walkers has been measured, separately. In our search of products and research to November 2026, we found no guide robot that shifts control in both directions from the user's own state (grip and following on the handle) together with familiarity of the route. Glide has a user-led mode and taught routes, but the user switches modes by hand and its help never adapts.

[1] RNIB key statistics · [2] RNIB "Life on Pause" FOI report, Oct 2025 · [3] Guide Dogs Annual Report 2025 · [4] RNIB rail report via Optometry Today, 22 Jan 2026 · [5] RAIB Eden Park report, 2021 · [6] RNIB "In My Way", 2025 · [7] Schöller et al., RA-L 2020 · [8] Kayukawa et al., IMWUT 2020 · [9] Trautman & Krause, IROS 2010. Competitor matrix and full sources in the project notes.